

MATH-1014 T10A

Tutorial 1

WANG Yixiao

Department of Mathematics

February 9, 2026



Table of Contents

① Review of Basic Rules

- Differentiation Rules
- Integration Rules

② Definite Integral

- Definition
- Several Properties

Differentiation Rules

Basic Rules:

- Power Rule: $\frac{d}{dx}[x^n] = nx^{n-1}$
- Exponential: $\frac{d}{dx}[e^x] = e^x$
- Natural Log: $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$

Advanced Rules:

- Product: $(uv)' = u'v + uv'$
- Quotient: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$
- Chain:
$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

Differentiation Rules

Problem 1: Find $\frac{d}{dx} \left[\frac{x \ln(x)}{e^x} \right]$

This problem tests:

- Product Rule
- Quotient Rule
- Natural Logarithm Differentiation
- Exponential Function Differentiation

Differentiation Rules

Problem 1: $\frac{d}{dx} \left[\frac{x \ln(x)}{e^x} \right]$

$$\begin{aligned} \frac{d}{dx} \left(\frac{x \ln(x)}{e^x} \right) &= \frac{(x \ln x)' e^x - (e^x)' (x \ln x)}{(e^x)^2} \\ &= \frac{(x \ln x)' - x \ln x}{e^x} \\ &= \frac{(\ln x + x \cdot \frac{1}{x}) - x \ln x}{e^x} \\ &= \frac{(1 - x) \ln x + 1}{e^x}. \end{aligned}$$

Integration Techniques

- **Basic Integrals:**

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x dx = e^x + C$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

- **Integration by Substitution:**

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

- **Integration by Parts:**

$$\int u dv = uv - \int v du$$

Integration Techniques

Problem 2: Evaluate $\int \frac{\ln(x)}{x} \sqrt{1 + \ln^2(x)} dx$

This problem tests:

- Integration by Substitution
- Recognition of Complex Patterns
- Chain Rule Understanding

Integration Techniques

Problem 2:

Let $u(x) = \ln x$

$$\begin{aligned} du &= \frac{1}{x} dx \\ \int \frac{\ln x}{x} \sqrt{1 + \ln^2 x} dx &= \int \frac{u(x)}{x} \sqrt{1 + u^2(x)} dx \\ &= \int u \sqrt{1 + u^2} du \\ &= \int u(1 + u^2)^{1/2} du \\ &= \frac{1}{3}(1 + u^2)^{3/2} + C \\ &= \frac{1}{3}(1 + \ln^2 x)^{3/2} + C. \end{aligned}$$

Problem 3

If $f(x) = 4 \ln(\ln(x))$, find $f'(x)$.

Problem 4

Let $y = \tan^{-1}(\sqrt{3x^2 - 1})$. Then $\frac{dy}{dx} =$

Identify the structure of the composite function first!

Solutions

Problem 3

If $f(x) = 4 \ln(\ln x)$, find $f'(x)$.

$$\text{Let } g(x) = \ln x, \quad f(x) = 4 \ln(g(x)).$$

$$\begin{aligned} f'(x) &= 4 \cdot \frac{1}{g(x)} g'(x) \\ &= 4 \cdot \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{4}{x \ln x}. \end{aligned}$$

Problem 4

Let $y = \tan^{-1}(\sqrt{3x^2 - 1})$. Find $\frac{dy}{dx}$.

$$y = \arctan u, \quad u = \sqrt{3x^2 - 1}.$$

$$\begin{aligned} y' &= \frac{1}{1 + u^2} u' \\ &= \frac{1}{1 + (3x^2 - 1)} \cdot \frac{6x}{2\sqrt{3x^2 - 1}} = \frac{1}{x\sqrt{3x^2 - 1}}. \end{aligned}$$

Integration by part

Problem 5

Evaluate $\int \frac{x}{\sqrt{x+5}} dx$.

Integration by substitution

Problem 6

Evaluate $\int \frac{\cos(t)}{(5 \sin(t) + 12)^2} dt$.

Solutions

Problem 5

Evaluate $\int \frac{x}{\sqrt{x+5}} dx$.

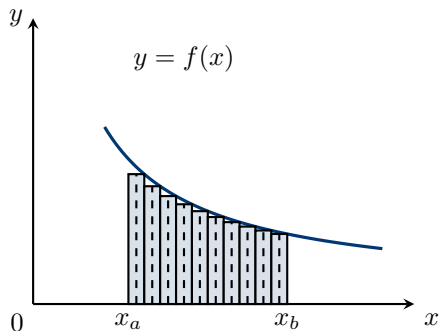
$$\begin{aligned}\int \frac{x}{\sqrt{x+5}} dx &= 2 \int x d\sqrt{x+5} = 2 \left(x\sqrt{x+5} - \int \sqrt{x+5} dx \right) \\ &= 2 \left(x\sqrt{x+5} - \frac{2}{3}(x+5)^{3/2} \right) + C.\end{aligned}$$

Problem 6

Evaluate $\int \frac{\cos(t)}{(5 \sin(t) + 12)^2} dt$. Let $u = \sin t$, $du = \cos t dt$

$$\begin{aligned}\int \frac{\cos t}{(5 \sin t + 12)^2} dt &= \int \frac{du}{(5u + 12)^2} \\ &= -\frac{1}{5(5u + 12)} + C = -\frac{1}{5(5 \sin t + 12)} + C.\end{aligned}$$

Definition of Definite Integral



$$\int_{x_a}^{x_b} f(x) dx$$

$$\text{Area} \approx \sum_{i=1}^n f(x_i^*) \Delta x_i$$

$$\text{Area} = \lim_{\max \Delta x_i \rightarrow 0} \sum_{i=1}^n f(x_i^*) \Delta x_i$$

Riemann sum!

Fundamental Theorem of Calculus

Part 1: If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Part 2: For continuous f ,

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

Problem 7 $F(x) = \int_0^x \sin(t^2) dt$. Find $F'(x)$.

Problem 8

Evaluate the integral $\int_0^4 (2e^x + 8 \cos x) dx$.

Try to find the antiderivative first!

Problem 9

Suppose that $F(x) = \int_1^x f(t) dt$, where

$$f(t) = \int_1^{t^2} \frac{\sqrt{6+u^6}}{u} du.$$

Find $F''(2)$.

Solutions

Problem 8

Evaluate $\int_0^4 (2e^x + 8 \cos x) dx$.

$$\begin{aligned}\int_0^4 (2e^x + 8 \cos x) dx &= [2e^x + 8 \sin x]_0^4 \\ &= 2e^4 + 8 \sin 4 - 2.\end{aligned}$$

Problem 9

Given $F(x) = \int_1^x f(t) dt$ and $f(t) = \int_1^{t^2} \frac{\sqrt{6+u^6}}{u} du$, find $F''(2)$.

$$F'(x) = f(x) \Rightarrow F''(x) = f'(x).$$

$$f'(t) = \frac{\sqrt{6+(t^2)^6}}{t^2} \cdot \frac{d}{dt}(t^2) = \frac{\sqrt{6+t^{12}}}{t^2} \cdot 2t = \frac{2\sqrt{6+t^{12}}}{t}.$$

$$F''(2) = f'(2) = \frac{2\sqrt{6+2^{12}}}{2} = \sqrt{6+4096} = \sqrt{4102}.$$

Problem 10

Evaluate the following limit by first recognizing the sum as a Riemann sum for a function defined on $[0, 1]$:

$$\lim_{n \rightarrow \infty} \frac{2}{n} \left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \sqrt{\frac{3}{n}} + \cdots + \sqrt{\frac{n}{n}} \right).$$

Recognize it as a Riemann sum on $[0, 1]$!

Solutions

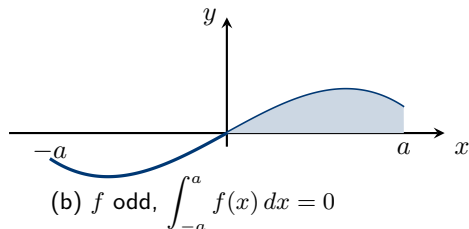
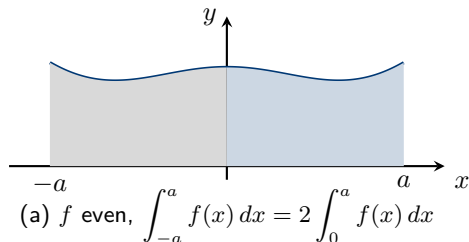
Problem 10

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n \sqrt{\frac{i}{n}} &= \lim_{n \rightarrow \infty} 2 \sum_{i=1}^n \sqrt{\frac{i}{n}} \cdot \frac{1}{n} \\ &= 2 \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x,\end{aligned}$$

$$\Delta x = \frac{1}{n}, \quad x_i = \frac{i}{n} \in [0, 1], \quad f(x) = \sqrt{x}$$

$$= 2 \int_0^1 \sqrt{x} \, dx = 2 \left[\frac{2}{3} x^{3/2} \right]_0^1 = \frac{4}{3}.$$

Symmetry



Note

- Even: $f(-x) = f(x)$
- Odd: $f(-x) = -f(x)$
- Use symmetry to simplify integrals on $[-a, a]$.

Problem 11 Calculate $I = \int_{-1}^1 \left(\ln(x + \sqrt{1+x^2}) + 2x + e^{-x} \right) dx$.

Solution

Problem

$$u(x) = \ln(x + \sqrt{1+x^2}).$$

$$\begin{aligned} u(-x) + u(x) &= \ln(-x + \sqrt{1+x^2}) + \ln(x + \sqrt{1+x^2}) \\ &= \ln((\sqrt{1+x^2} - x)(\sqrt{1+x^2} + x)) = \ln(1) = 0. \end{aligned}$$

Hence $u(x)$ is odd, so $\int_{-1}^1 u(x) dx = 0$. Therefore

$$\begin{aligned} I &= \int_{-1}^1 (u(x) + (2x+1)e^{-x}) dx = \int_{-1}^1 (2x+1)e^{-x} dx \\ &= \int_{-1}^1 (2x+1) d(e^{-x}) = -[(2x+1)e^{-x}]_{-1}^1 + \int_{-1}^1 2e^{-x} dx \\ &= -(3e^{-1} - (-1)e^1) + [-2e^{-x}]_{-1}^1 = e - \frac{5}{e}. \end{aligned}$$

THANK YOU!

